

# Welcome to the Foothills of Crude's Next Multiyear Boom Cycle

**Bob McNally**

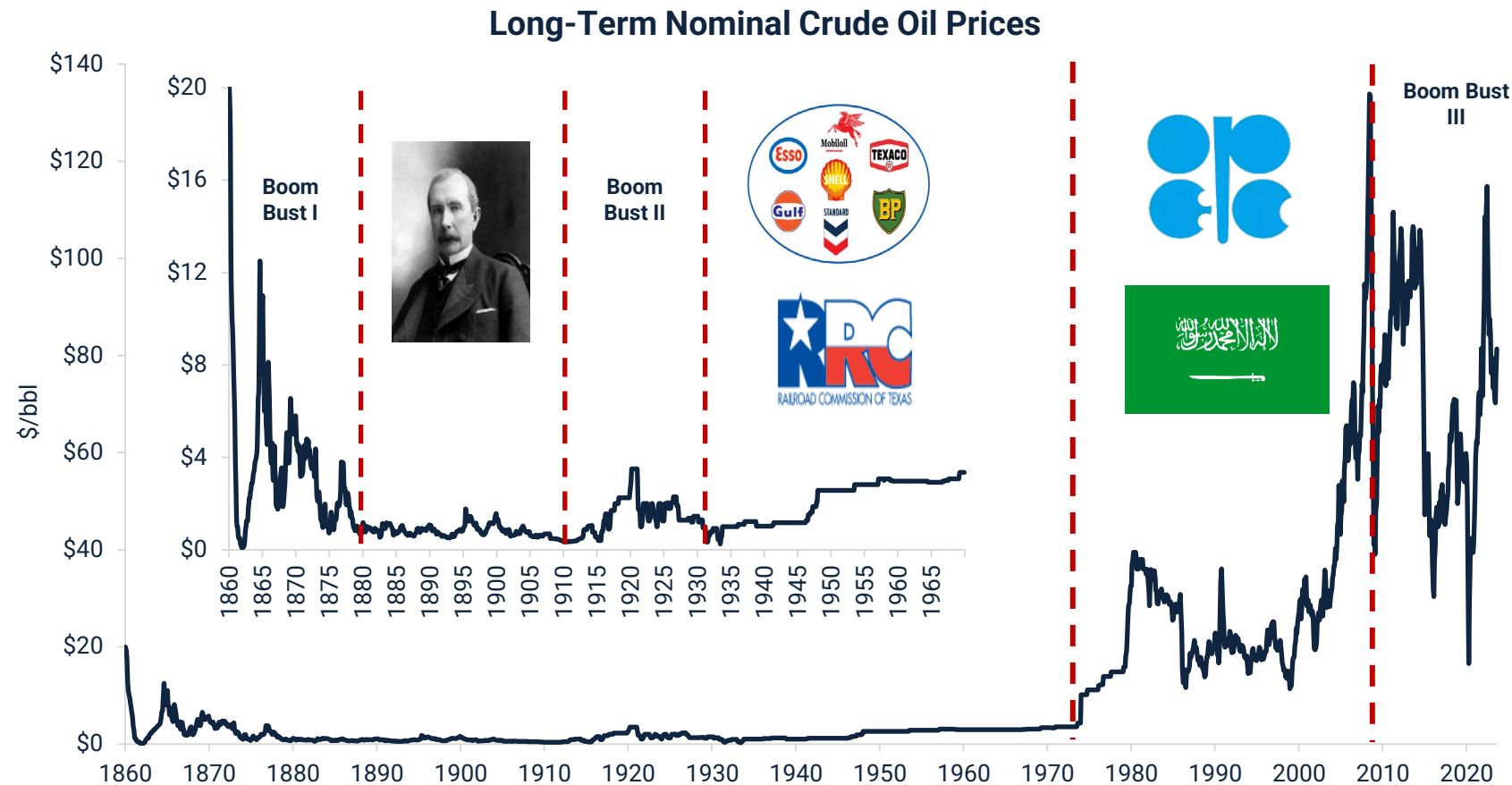
**Rapidan Energy Group**

Presentation to EnCap Investments, L.P.

November 6, 2023

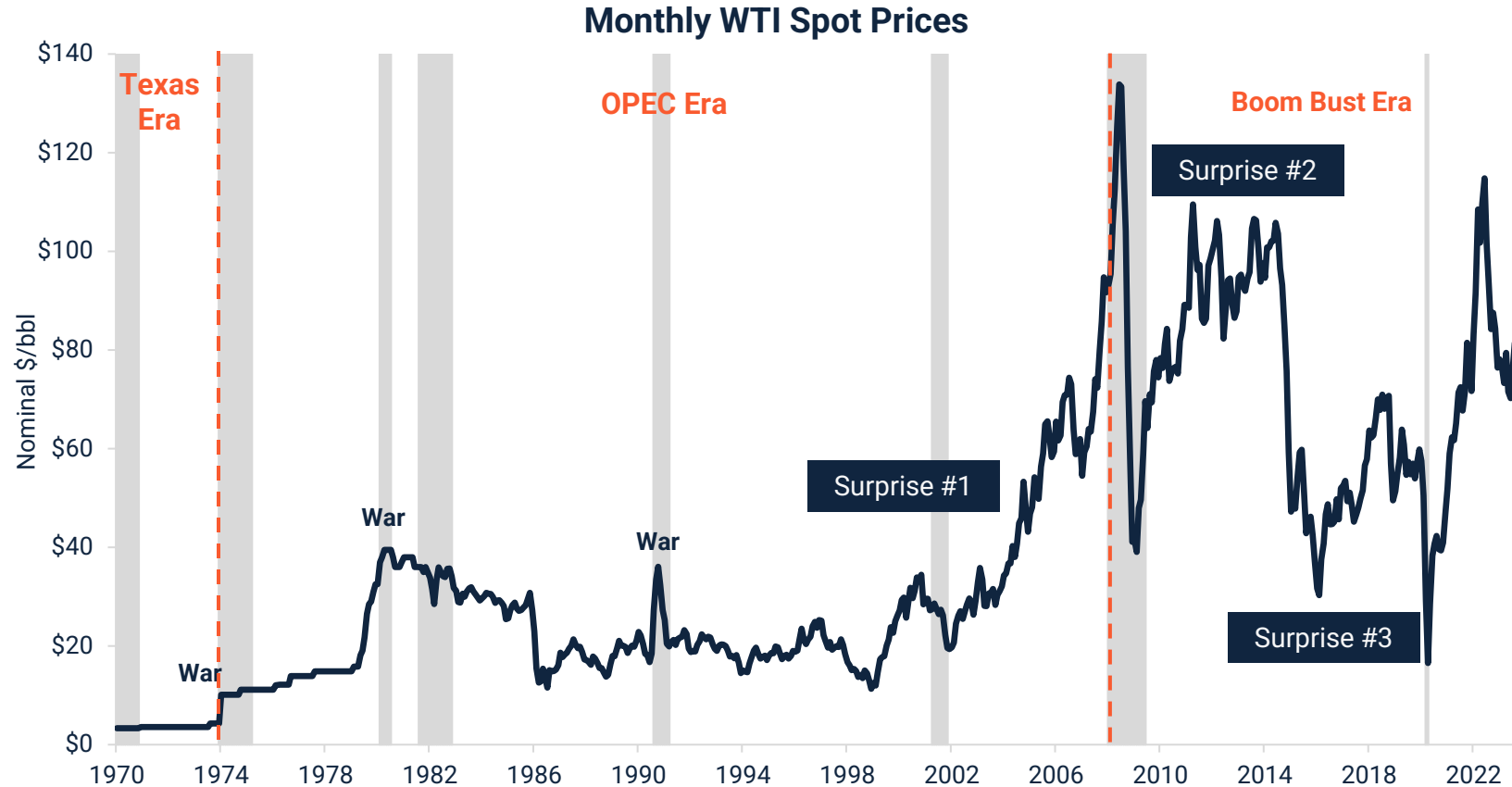
Houston, Texas

# 51 Years Ago (Not a Typo) OPEC Took Over From the Texas Railroad Commission



Source: Rapidan Energy Group, The Derrick, API, St. Louis Fed, EIA

# The Last Three Big Surprises Came on the Supply Side



Note: Grey shading represents a recession in the US, as defined by the National Bureau of Economic Research (NBER).  
Source: Rapidan Energy Group, The Derrick, API, St. Louis Fed, EIA

Confidential  
Sam Sam  
Jun 26, 2024 12:06 PM EDT

# Disclaimer: I Am Not an Evil Person Who Hates the Environment

**Net-Zero Footprint 2 Years  
Peace Corps: 1988-1990**



**Early Adopter of Hybrid Technology:  
2007**



**White House Energy Advisor:  
2001-2003**

- Unfroze and reformed fuel economy standards
- Residential solar tax credit
- Fusion energy test reactor
- Hydrogen Fuel Initiative



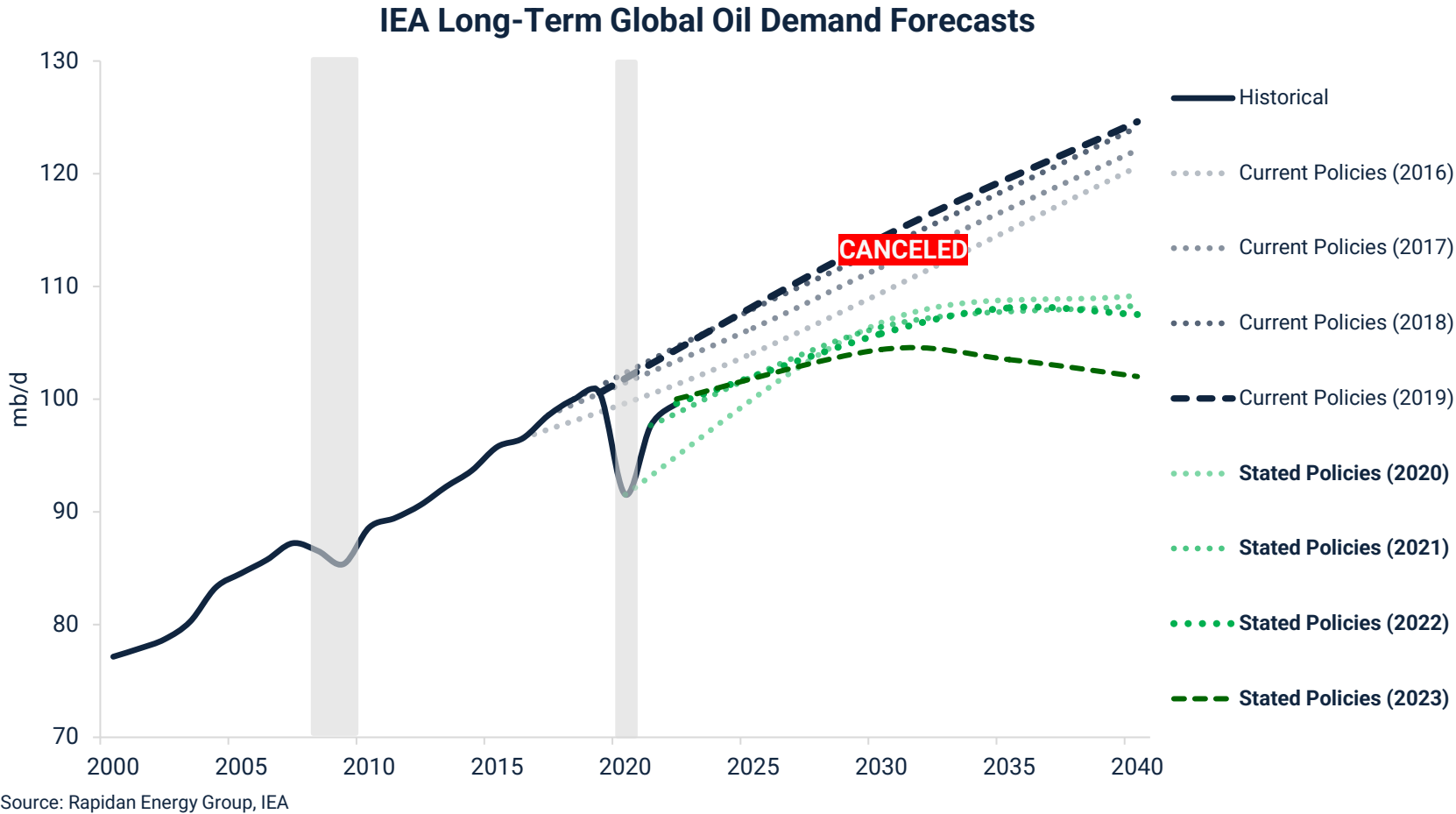


# The Next Big Surprise: Peak Oil Demand Proves to be Wishful Thinking

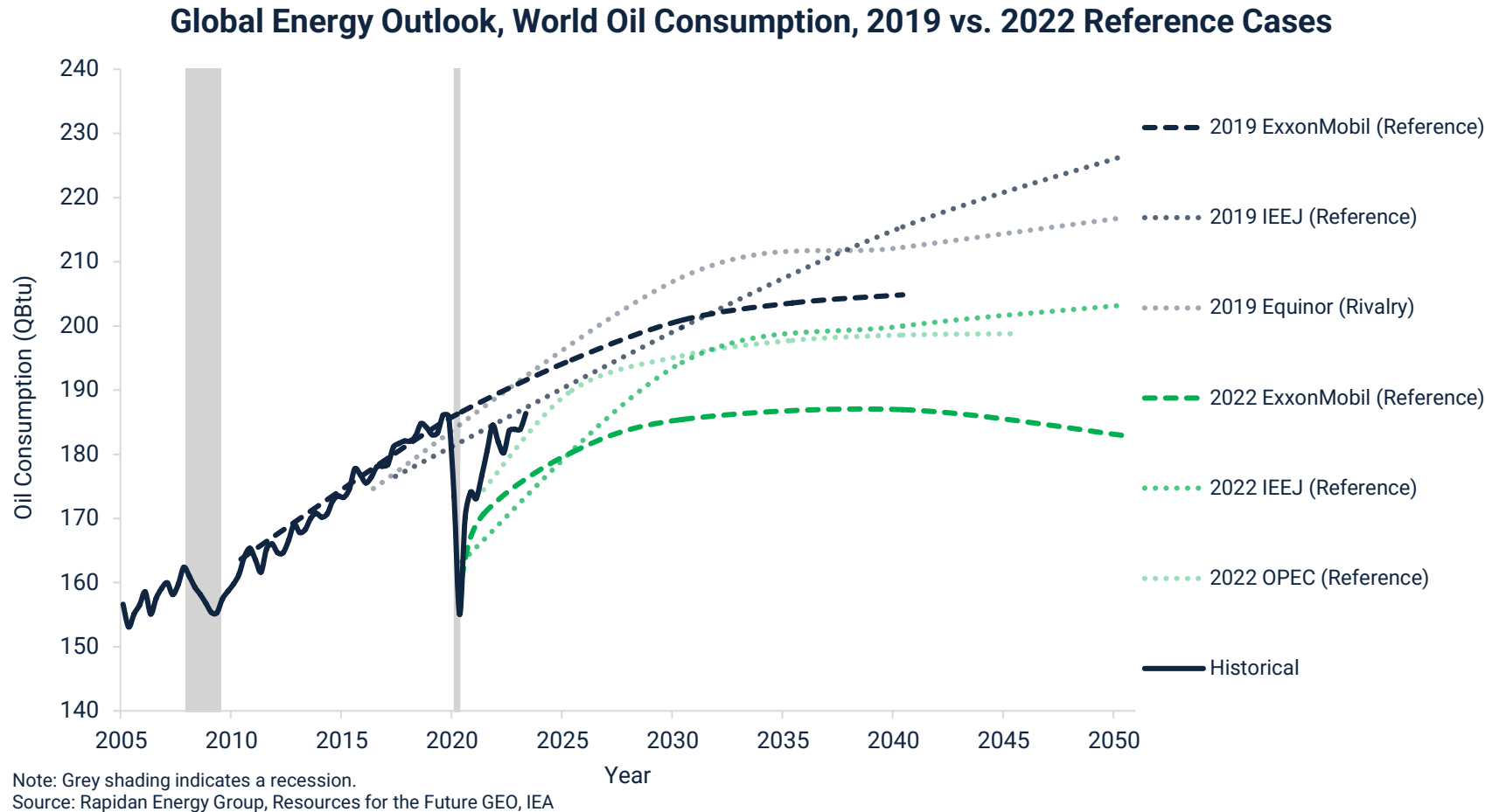
**The Guardian** Thu 5 Apr 2018  
**IEA accused of undermining global shift from fossil fuels**  
Highly critical study warns projections used by the organisation tasked with leading the switch to clean energy remain skewed towards oil and gas and may break climate targets of Paris agreement

**OILCHANGE INTERNATIONAL**  
**IEA dogged by criticism at UN climate talks**  
DECEMBER 17, 2019 | BY NATHAN LEMPHERS | BLOG POST, ENERGY TRANSITIONS & FUTURES

**FINANCIAL TIMES** APRIL 2 2019  
**IEA's climate models criticised as too fossil-fuel friendly**  
Investors say body's benchmark energy forecasts are not in line with latest science

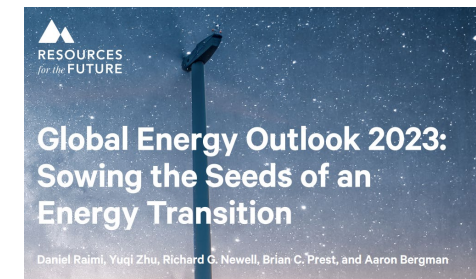
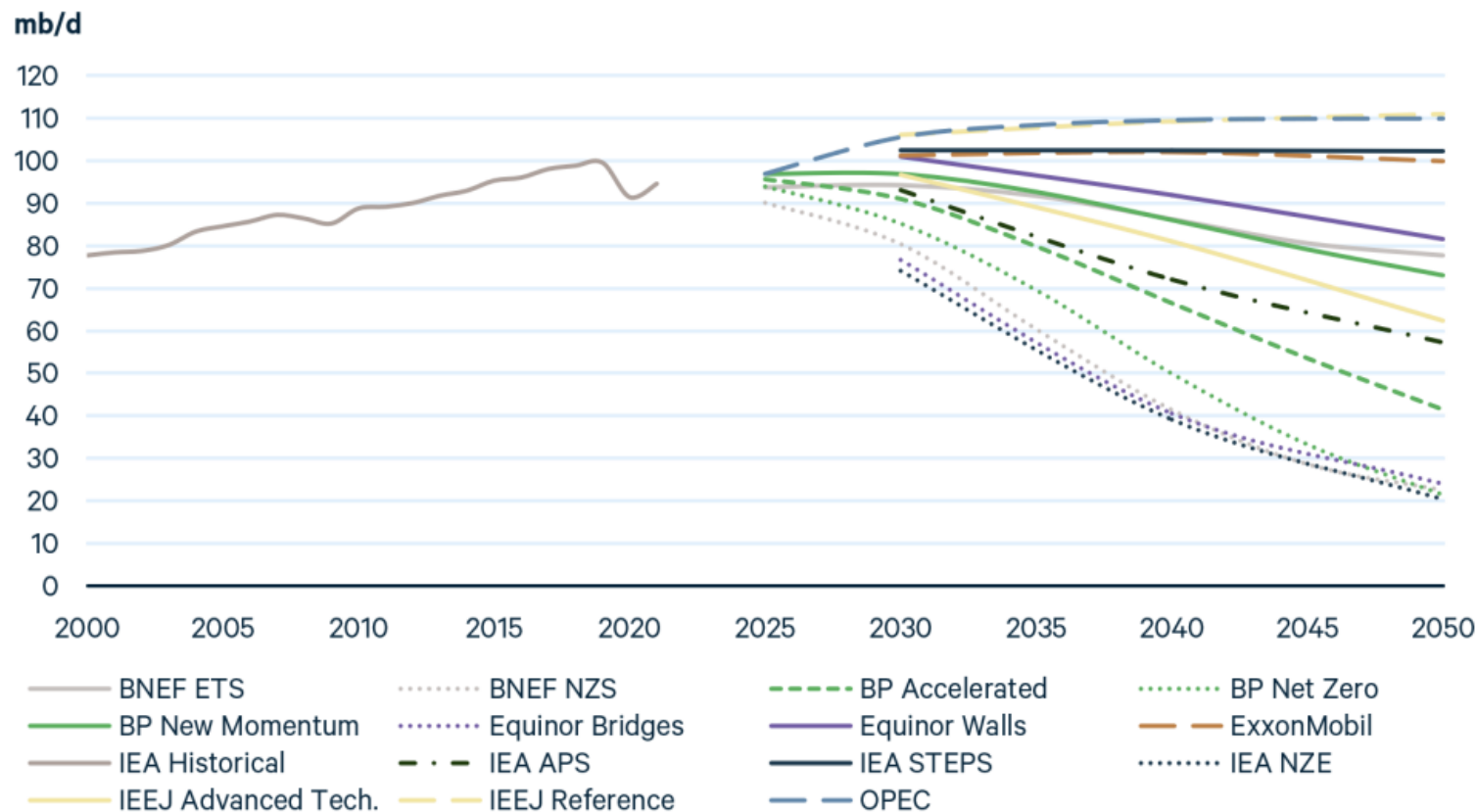


# The Consensus Followed the IEA into Imaginary Peak Oil Demand



# Consensus Now Expects Oil Demand to Peak or Collapse Soon

Figure 8. World Oil Demand



# Underappreciated Risk – Public Blowback Over Higher Energy Costs

Rapidan’s Decarbonization Blowback Tracker follows instances where policy-induced fuel price increases encounter grassroots opposition in the form of public protests.

Larger oil consuming countries (demand >450 kb/d) are often unwilling to bear the political price of sustained price increases that face public resistance.

Austria, France, Chile, and Mexico have all reversed course on fuel price increases after protests.

In Iran in 2019, officials refused to reverse their policies and ended up killing at least 200 protestors.

Smaller, poorer consumers (ex. Haiti, Lebanon, and Bangladesh) are often forced to reduce oil subsidies to control debt.

These countries often can’t afford to reverse these price increases and, as a result, face sustained instability.

Politicians prefer non-binding targets that lead to less demand destruction versus enforceable legislation that could imperil their political futures.

| Decarbonization Blowback Tracker |                        |                                                                                                                                                                                                                                                                                                                  |                                                                                                             |
|----------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Country                          | 2022 Oil Demand (kb/d) | Protest Description                                                                                                                                                                                                                                                                                              | Policy Change Reversed?                                                                                     |
| OECD                             |                        |                                                                                                                                                                                                                                                                                                                  |                                                                                                             |
| Austria<br>September 2022        | 250                    | Citizens protested against high energy prices and related EU sanctions spurred by the Russian invasion of Ukraine and climate-forward policies.                                                                                                                                                                  | Yes, Austria delayed its €30/tonne carbon tax.                                                              |
| Chile<br>October 2019            | 380                    | Chilean students lead massive protests against 4% increase in subway fares, which the government said was necessary to cope with rising oil prices and currency weakness. Protests caused the government to relocate COP25 from Santiago to Madrid, Spain.                                                       | Yes. Economic reforms and constitutional referendum.                                                        |
| France<br>October 2018           | 1,600                  | "Yellow Vest" protests erupt over carbon taxes on fuel.                                                                                                                                                                                                                                                          | Yes. Reversal of fuel price increases.                                                                      |
| Mexico<br>January 2017           | 1,900                  | Protests break out after the government's decision to deregulate gasoline prices caused pump prices to rise 20%. Protests hit at a time when the peso's value was plummeting and inflation was rising. After the increases gasoline was \$2.85/gallon.                                                           | Yes. Election of AMLO and reversal of deregulated gasoline prices.                                          |
| Non-OECD                         |                        |                                                                                                                                                                                                                                                                                                                  |                                                                                                             |
| Bangladesh<br>November 2022      | 200                    | Truckers and opposition political parties protested and passenger transit workers called for a strike after the government increased fuel prices by 23%.                                                                                                                                                         | No                                                                                                          |
| Haiti<br>September 2022          | 20                     | Following IMF recommendations, the Haitian president reduced eliminated fuel subsidies-- doubling its price.                                                                                                                                                                                                     | No                                                                                                          |
| Morocco<br>June 2022             | 290                    | Morocco successfully ended its fuel subsidy scheme in 2015, but high global energy prices have spurred protests demanding the subsidies be reinstated.                                                                                                                                                           | Yes, they implemented a fuel aid mechanism for transport operators.                                         |
| South Africa<br>June 2022        | 570                    | Protests erupted after the government allowed temporary suspension of fuel levies to expire.                                                                                                                                                                                                                     | No, the government left the levies.                                                                         |
| Kazakhstan<br>January 2022       | 330                    | Protests erupted after the government removed price caps on LPG, increasing prices by 100% .                                                                                                                                                                                                                     | Yes; The President reinstituted price caps on LPG at 50 KZT/L, and the Prime Minister and cabinet resigned. |
| Thailand<br>October 2021         | 1,490                  | <b>November 2021:</b> LTFT continues protests - government caps prices at 28 THB/L, a 7% reduction from October<br><b>October 2021:</b> Truckers' organization, Land Transport Federation of Thailand, protests in October asking the government to cap diesel prices at 25 THB /L. Government caps at 30 THB/L. | Yes; Government has adjusted the cap but has not yet met protestor demands.                                 |
| Lebanon<br>June 2021             | 150                    | <b>Aug 12:</b> Protestors shut down streets after Lebanese Energy Minister hikes fuel prices by more than 30% because central bank runs low on Forex reserves.                                                                                                                                                   | No                                                                                                          |
| Sudan<br>June 2021               | 130                    | Sudanese Prof. Association (labor group which ousted former dictator Omar al-Bashir) called for daily protests after the government increased fuel prices \$~0.35/liter as part of IMF reforms.                                                                                                                  | No                                                                                                          |
| Nepal<br>February 2021           | 60                     | All Nepal National Free Student Union protested in Kathmandu over the Nepal Oil Corp. decision to raise price of petrol and cooking gas.                                                                                                                                                                         | No                                                                                                          |
| Iran<br>November 2019            | 1,800                  | Fuel subsidy cuts spark widespread demonstrations and violent backlash by security forces. The government estimates ~200 people were killed.                                                                                                                                                                     | No.                                                                                                         |





# Energy Policy Post-Ukraine: Subsidize Fuel, Delay on Climate, Grab Profits

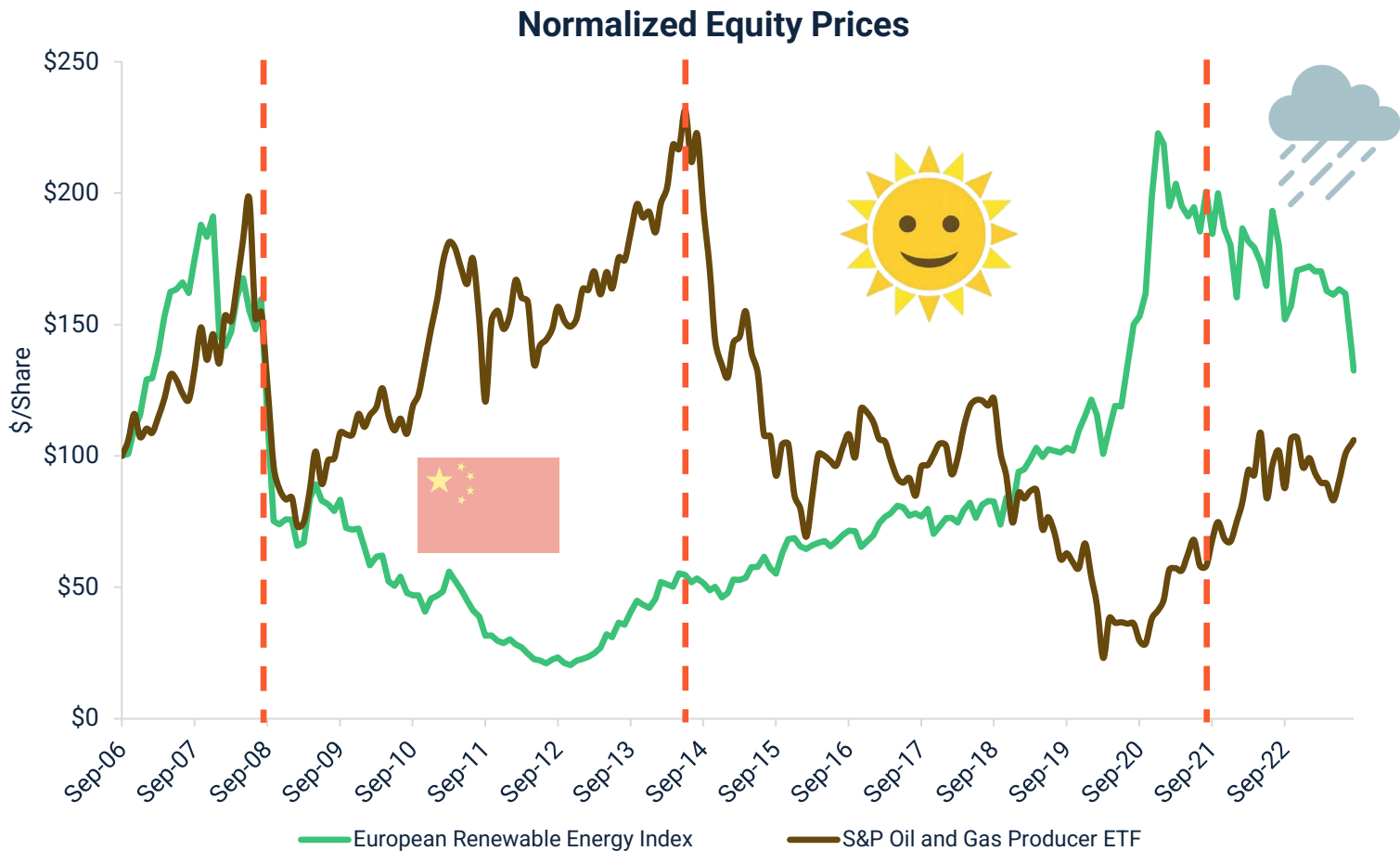
Political will to impose climate policies evaporates when energy costs rise

By February of 2023, Western governments – largely European – allocated \$750 billion to direct stimulus, tax cuts, subsidies, and corporate bailouts, hoping to lessen the bite of soaring energy costs on their citizenry.

- 23 governments subsidized or lowered taxes on gasoline, diesel, natural gas, and electricity – making this the most popular policy response to increased energy costs.
  - Many of these provisions expired in mid-2023, but extensions and renewals are likely, especially if inflation persists and a global recession begins – for example, Germany’s recently proposed energy subsidy for its industrial base.
- 21 governments issued direct stimulus to households to help with monthly bills and pump prices.
- 15 governments (exclusively in Europe) passed windfall profits taxes. **The EU led the way with a bloc-wide tax aimed at surplus profits of both fossil fuel companies and low-cost power producers like nuclear and renewable energy.**
- 6 governments rolled-back or delayed existing climate policies like renewable fuel obligations, levies to fund renewables deployment, GHG targets and carbon taxes, and EV subsidies.
- Countries with oil and gas stockpiles (like the US’s Strategic Petroleum Reserve) have tapped those resources. 2 governments – Hungary and Latvia – opted to restrict exports of refined products.

| Government     | Energy Market Interventions      |                               |                      |                              |                 |
|----------------|----------------------------------|-------------------------------|----------------------|------------------------------|-----------------|
|                | Fuel Subsidies or Tax Reductions | Direct Stimulus to Households | Windfall Profits Tax | Rollback of Climate Policies | Export Controls |
| European Union |                                  |                               | x                    |                              |                 |
| Austria        | x                                | x                             | x                    | x                            |                 |
| Belgium        | x                                | x                             |                      |                              |                 |
| Bulgaria       | x                                |                               | x                    |                              |                 |
| Croatia        | x                                | x                             |                      |                              |                 |
| Cyprus         |                                  |                               | x                    |                              |                 |
| Czech Republic |                                  | x                             | x                    | x                            |                 |
| Denmark        |                                  | x                             |                      |                              |                 |
| Estonia        |                                  |                               |                      |                              |                 |
| Finland        | x                                |                               |                      | x                            |                 |
| France         | x                                | x                             |                      |                              |                 |
| Germany        | x                                | x                             | x                    | x                            |                 |
| Greece         |                                  | x                             | x                    |                              |                 |
| Hungary        | x                                |                               | x                    |                              | x               |
| Ireland        |                                  | x                             | x                    |                              |                 |
| Italy          | x                                | x                             | x                    |                              |                 |
| Latvia         | x                                | x                             |                      |                              | x               |
| Lithuania      | x                                | x                             |                      |                              |                 |
| Luxembourg     |                                  | x                             |                      |                              |                 |
| Malta          | x                                | x                             |                      |                              |                 |
| Netherlands    | x                                |                               | x                    |                              |                 |
| Poland         | x                                | x                             |                      |                              |                 |
| Portugal       | x                                |                               |                      | x                            |                 |
| Romania        |                                  | x                             | x                    |                              |                 |
| Slovakia       |                                  |                               | x                    |                              |                 |
| Slovenia       |                                  | x                             |                      |                              |                 |
| Spain          | x                                | x                             |                      |                              |                 |
| Sweden         | x                                | x                             |                      | x                            |                 |
| Norway         |                                  |                               | x                    |                              |                 |
| United Kingdom | x                                | x                             | x                    |                              |                 |
| Canada         |                                  |                               |                      |                              |                 |
| United States  |                                  |                               |                      |                              |                 |
| California     | x                                | x                             |                      |                              |                 |
| Connecticut    | x                                |                               |                      |                              |                 |
| Florida        | x                                |                               |                      |                              |                 |
| Georgia        | x                                |                               |                      |                              |                 |
| Hawaii         | x                                |                               |                      |                              |                 |
| Maryland       | x                                |                               |                      |                              |                 |
| New York       | x                                |                               |                      |                              |                 |
| Totals         | 25                               | 21                            | 15                   | 6                            | 2               |

# Beware of Status Quo Bias: Atmospheric Conditions are Changing



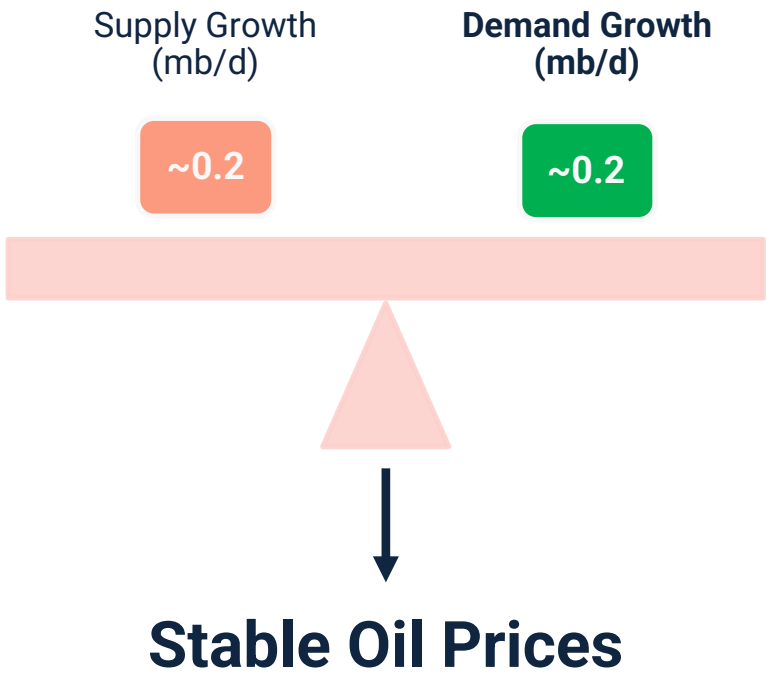
|                | 2015-2020<br>Paris-to-Biden | 2021-?<br>Delay & Distraction |
|----------------|-----------------------------|-------------------------------|
| Oil prices     | Low                         | High                          |
| Interest Rates | Low                         | High                          |
| Inflation      | Low                         | High                          |
| China          | Partner                     | Adversary                     |
| Russia         | Peace                       | War                           |



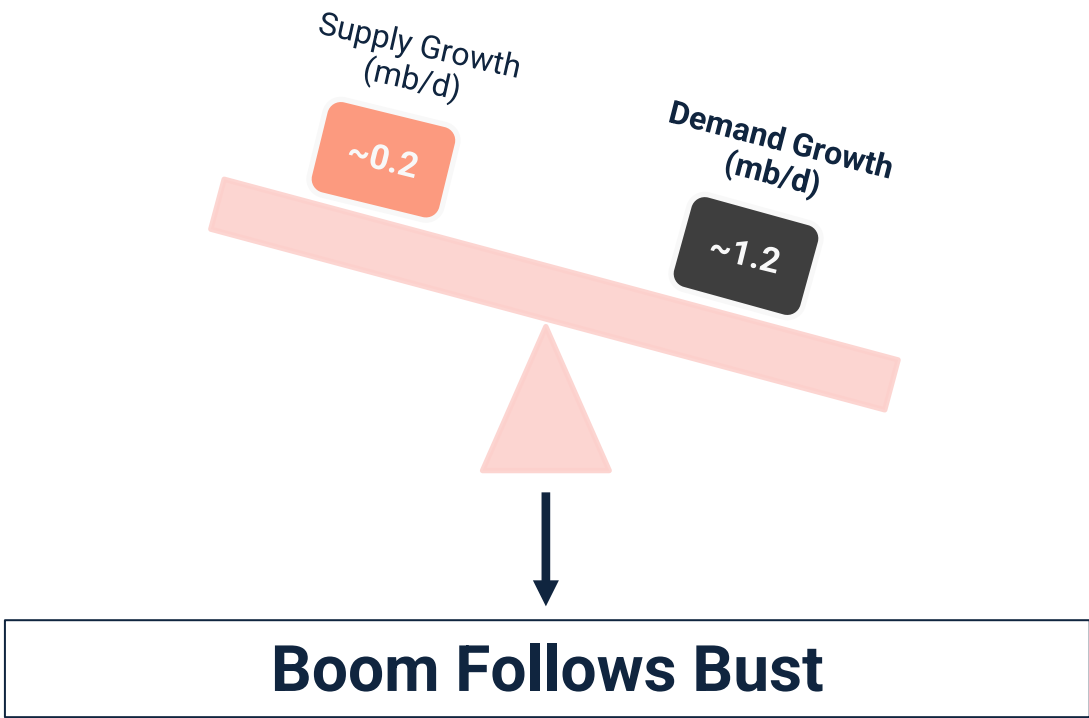
|                                                   |
|---------------------------------------------------|
| 2009-2014<br>Build It In China and They Will Come |
| EV, Solar, Wind Supply Chains                     |

# When Consensus Capitulates to Stronger Demand, Boom Cycle Erupts

**Consensus View:**  
Peak Oil Demand is Near, So  
Underinvestment Not a Problem



**Rapidan Energy Group's View:**  
Peak Oil Demand is Wishful Thinking,  
Underinvestment is a Problem, Buckle Up



Note: Consensus view reflects the IEA's average global demand and supply growth between 2022-2035 in its 2023 *World Energy Outlook (WEO)*.

# Thank You!

Bob.McNally@RapidanEnergy.Com